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NMR Spectral Component Estimation using Unbalanced Optimal Transport via Mirror Descent Algorithm

Bachelor's thesis
in MATHEMATICS

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Abstract

This thesis studies a scalable approach to quantitative NMR mixture analysis using unbalanced optimal transport—a differentiable objective combining transport cost with regularization terms that allow partial mass transport, via mirror descent. We build on the Magnetstein method [DNG⁺23], which formulates spectral comparison as a linear program under the Wasserstein distance, but suffers from high computational cost. To improve efficiency, we impose a multidiagonal sparsity structure on the transport plan, reducing memory usage from quadratic to linear in the resolution. Our implementation extends the Python Optimal Transport [FCGea21] library and includes support for custom regularizers such as total variation. Experimental results on synthetic and real NMR spectra show that our method retains the robustness of Wasserstein-based comparisons while offering substantial gains in scalability.

Keywords

Wasserstein, Mirror Descent, Optimal Transport, Python, Spectroscopy, Nuclear Magnetic Resonance

Thesis domain (Socrates-Erasmus subject area codes)

11.1 Matematyka

Subject classification

CCS

- Applied computing
 - Physical sciences and engineering
 - Chemistry
 - Mathematics and statistics
- Computing methodologies
 - Machine learning
 - Machine learning algorithms

Tytuł pracy w języku polskim

Estymacja składników mieszanin w spektroskopii NMR z wykorzystaniem niezbilansowanego optymalnego transportu przy użyciu algorytmu mirror descent

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Chapter 1

Introduction

In their recent work, researchers from the Faculty of Mathematics, Informatics and Mechanics of the University of Warsaw (MIMUW) introduced *Magnetstein* [DNG⁺23], an open-source tool for quantitative NMR mixture analysis that applies optimal transport theory to address challenges such as low resolution, peak overlap, and signal distortion. By representing spectra as probability measures and comparing them using the Wasserstein distance, the method enables robust quantification even under adverse experimental conditions. While highly accurate, the approach is computationally expensive: it formulates the problem as a linear program solved via the simplex method and relies heavily on commercial solvers such as *Gurobi*. This makes it difficult to scale to large datasets or high-dimensional spectra, where linear programming becomes increasingly inefficient and memory-intensive, and its operation depends on changes in external libraries, which complicates long-term maintainability.

These limitations motivate the exploration of alternative optimization techniques that retain the geometric advantages of Wasserstein-based comparison while improving scalability. In this thesis, we investigate a solution based on mirror descent — a first-order optimization method that operates in non-Euclidean geometries. This approach introduces a noise-tolerant formulation that plays a similar role to the κ -mixture and κ -component “trashcan” regularization used in *Magnetstein*, allowing the method to account for outliers and model uncertainty. Our aim is to reduce computational overhead while maintaining robustness to peak shifts and unresolved signal contributions. The core of our implementation builds upon the Python Optimal Transport library [FCG_{ea}21], which we adapt and extend to efficiently recover mixture proportions under a Wasserstein-inspired loss, broadening the applicability of transport-based spectral analysis in practice.

Chapter 2

Nuclear Magnetic Resonance Spectra

Nuclear Magnetic Resonance (NMR) spectroscopy is a fundamental technique in analytical chemistry and structural biology, offering non-destructive insights into the composition of molecular mixtures. It operates by measuring the response of atomic nuclei—typically ^1H or ^{13}C —to a strong magnetic field and a sequence of radiofrequency pulses.

These nuclei, when placed in such a field, exhibit resonance frequencies that depend on their electronic structure, which allows us to interpret the NMR spectrum as a unique chemical fingerprint of a substance. The resulting spectra encode characteristic signals (peaks) whose positions and intensities carry information about molecular structure and concentration. A representative visualization is presented in Figure 2.1.

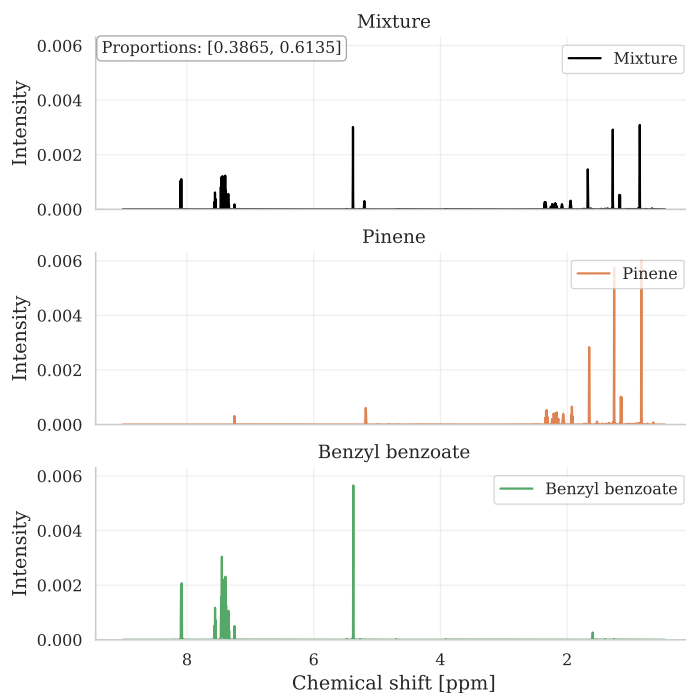


Figure 2.1: NMR mixture (top) and component spectra.

A central objective in quantitative NMR mixture analysis is to estimate the relative concentrations of known reference components that best reconstruct an observed mixture spectrum. Formally, given a set of m component spectra $S_1, \dots, S_m$ and a mixture spectrum S_{mix} , the goal is to recover nonnegative weights $w_1, \dots, w_m$ such that their linear

combination approximates the mixture:

$$S_{\text{mix}} \approx \sum_{i=1}^m w_i S_i. \quad (2.1)$$

Practically, NMR data is represented as a discrete set of measured intensity values at specific chemical shifts (pairs chemical shift, intensity). Observed spectra frequently exhibit peak shifts and noise due to instrumental imperfections, solvent interactions, and limited resolution. Such factors make direct peak matching ineffective for mixture deconvolution and force us to seek alternative, noise-tolerant methods.

In this thesis, we model NMR spectra as discrete probability distributions over the chemical shift axis. To apply probability-based metrics, intensities are normalized. This normalization is purely technical; we emphasize that no randomness assumption is implied. Instead, it enables the application of robust statistical distances, such as the Wasserstein distance, and ensures that a convex combination of spectra yields a valid spectrum—simplifying quantitative analysis.

2.1. Wasserstein Distance

The Wasserstein distance, also known as the Earth Mover’s Distance, quantifies the dissimilarity between two probability distributions. Intuitively, it represents the minimal “cost” required to transform one distribution into another by optimally “moving mass” across a given metric space.

Formally, let μ and ν be two probability distributions defined on a metric space $(\mathcal{X}, d)$. The p -Wasserstein distance between them is defined as:

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p}$$

where $\Pi(\mu, \nu)$ denotes the set of all joint distributions on $\mathcal{X} \times \mathcal{X}$ with marginals μ and ν . The function $d(x, y)$ represents the cost of moving a unit mass from x to y , and $p \geq 1$ is the order of the distance.

In practical scenarios, such as ours, we often work with discrete probability distributions. Suppose we have two discrete distributions represented as weighted sums of Dirac measures:

$$\mu = \sum_{i=1}^n a_i \delta_{x_i}, \quad \nu = \sum_{j=1}^m b_j \delta_{y_j}$$

with nonnegative weights a_i, b_j summing to 1, and support points $\{x_i\}$ and $\{y_j\}$ in $\mathcal{X}$. The discrete formulation of the p -Wasserstein distance is then given by the solution to the following linear optimization problem:

$$W_p(\mu, \nu) = \left(\min_{T \in U(a, b)} \sum_{i=1}^n \sum_{j=1}^m T_{ij} d(x_i, y_j)^p \right)^{1/p}, \quad (2.2)$$

where $U(a, b)$ is the set of all transport plans:

$$U(a, b) = \{T \in \mathbb{R}_+^{n \times m} : T \mathbf{1}_m = a, T^\top \mathbf{1}_n = b\}. \quad (2.3)$$

In the Magnetstein framework [DNG⁺23], NMR spectra are represented as discrete probability measures and compared via optimal transport using the Wasserstein distance. This strategy matches entire spectral distributions rather than isolated peaks, yielding robustness against peak shifts, overlap, and missing signals. To account for unexplained signals—such as experimental noise or unknown chemical entities—additional auxiliary states are introduced to absorb unmatched mass, ensuring accurate estimation of component proportions even when the mixture spectrum deviates from the convex hull of the reference spectra.

In this thesis, we leverage this Wasserstein-based spectral representation and replace Magnetstein’s linear programming with an optimization approach using mirror descent. This change preserves the geometric robustness of Wasserstein distances while aiming to improve computational performance.

Chapter 3

Solving Optimal Transport via Mirror Descent

In this chapter, we define a convex objective that quantifies the “distance” between an estimated convex combination of spectra and an observed mixture. Minimizing this objective is our ultimate goal, as its minimizer should correspond to the best possible estimate of the true spectral composition.

3.1. Unbalanced Optimal Transport vs Wasserstein distance

As shown in Equation 2.2, the p -Wasserstein distance can be computed via a linear optimization problem. If we set $p = 1$ it transforms to the following:

$$W_p(\mu, \nu) = \min_{\gamma \in U(a,b)} \langle \gamma, \mathbf{M} \rangle_F$$

where $\mathbf{M}_{i,j} = d(x_i, y_j)$ and $\langle \gamma, \mathbf{M} \rangle_F = \sum_{i,j} \gamma_{ij} M_{ij}$ denote the Frobenius inner product between matrices γ and $\mathbf{M}$.

Calculating the distance is equivalent to finding the transport plan γ , which is where the name *Optimal Transport* originates. The *unbalanced* version arises when we drop the constraints in Equation 2.3 and instead introduce penalties for deviations — when $\gamma \mathbf{1}$ differs from a and $\mathbf{1}^T \gamma$ differs from b . These penalties ensure that γ still resembles a joint distribution.

3.2. Optimization Objective

We consider the unbalanced optimal transport problem with regularization over the transport plan and divergence penalties on the marginals. The goal is to compute the optimal transport plan γ by minimizing the following regularized objective:

$$\gamma^* = \arg \min_{\gamma \geq 0} (\langle \gamma, \mathbf{M} \rangle_F + \lambda \cdot \text{div}_{\text{reg}}(\gamma, \mathbf{c}) + \tau_1 \cdot \text{div}_m(\gamma \mathbf{1}, \mathbf{a}) + \tau_2 \cdot \text{div}_m(\gamma^T \mathbf{1}, \mathbf{b})) \quad (3.1)$$

where:

- $\mathbf{M} \in \mathbb{R}^{n \times m}$ is the ground cost matrix,
- $\gamma \in \mathbb{R}_+^{n \times m}$ is the transport plan,
- $\mathbf{a} \in \mathbb{R}_+^n$, $\mathbf{b} \in \mathbb{R}_+^m$ are source and target marginals,

- $\mathbf{c} \in \mathbb{R}_+^{n \times m}$ is a reference plan used only for regularization,
- $\text{div}_{\text{reg}}(\gamma, \mathbf{c})$ is a regularization term measuring divergence between transport plans (typically Kullback–Leibler)
- div_m is a marginal divergence penalizing deviation from the prescribed marginals (e.g., ℓ_2 , Kullback–Leibler, or total variation (TV)),
- λ , τ_1 , and τ_2 are hyperparameters controlling the strength of the corresponding penalties.

This formulation enables partial mass transport by relaxing the marginal constraints and penalizing their deviation from the targets. We use TV as the default marginal divergence, defined as:

$$\text{TV}(u, v) = \|u - v\|_1.$$

This choice promotes sharper cuts, allowing the model to remove entire peaks rather than just parts of them.

The problem’s geometry involves non-differentiable points, which makes gradient computation challenging. To address this, we include the regularized divergence term $\text{div}_{\text{reg}}(\gamma, \mathbf{c})$ in the objective to smooth the optimization. In our approach, we use the Kullback–Leibler divergence with mass conservation term:

$$\text{KL}(\gamma, \mathbf{c}) = \gamma \cdot \log\left(\frac{\gamma}{\mathbf{c}}\right) + (\mathbf{c} - \gamma).$$

3.3. Mirror Descent

This section is hugely inspired by [Bec17]

In convex optimization problems with constraints and possibly non-differentiable objectives, the *projected subgradient* method provides a general framework. It extends projected gradient descent by replacing gradients with subgradients, which always exist for convex functions.

Definition 3.3.1 (Projected Subgradient Descent). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function and let $C \subseteq \mathbb{R}^n$ be a closed convex set. The projected subgradient descent method generates a sequence $(x^k)_{k \geq 0}$ according to:*

$$x^{k+1} = \Pi_C(x^k - t_k g_k),$$

where $g_k \in \partial f(x^k)$ is a subgradient of f at x^k , $t_k > 0$ is the step size, and Π_C denotes the orthogonal projection onto C :

$$\Pi_C(z) = \arg \min_{x \in C} \|x - z\|^2.$$

This scheme can be interpreted as minimizing a local linear approximation of the function, regularized by a quadratic penalty. This corresponds to solving:

$$x^{k+1} = \arg \min_{x \in C} \left\{ f(x^k) + \langle \nabla f(x^k), x - x^k \rangle + \frac{1}{2t_k} \|x - x^k\|^2 \right\}.$$

Specifically, the linearized objective with quadratic regularization satisfies the identity:

$$f(x^k) + \langle \nabla f(x^k), x - x^k \rangle + \frac{1}{2t_k} \|x - x^k\|^2 = \frac{1}{2t_k} \left\| x - \left(x^k - t_k \nabla f(x^k) \right) \right\|^2 + D,$$

for some constant D independent of x .

Proof. Applying the identity $\|a + b\|^2 = \|a\|^2 + 2\langle a, b \rangle + \|b\|^2$, we get:

$$\|x - (x^k - t_k \nabla f(x^k))\|^2 = \|x - x^k\|^2 + 2t_k \langle \nabla f(x^k), x - x^k \rangle + t_k^2 \|\nabla f(x^k)\|^2.$$

Since the term $t_k^2 \|\nabla f(x^k)\|^2$ is independent of x , it may be omitted when taking the minimizer. Dividing the remaining expression by $2t_k$ (which does not affect the argmin) gives

$$\arg \min_{x \in C} \left\{ \langle \nabla f(x^k), x - x^k \rangle + \frac{1}{2t_k} \|x - x^k\|^2 \right\}.$$

Finally, adding the constant $\langle \nabla f(x^k), x^k \rangle$ yields exactly

$$\arg \min_{x \in C} \left\{ \langle \nabla f(x^k), x \rangle + \frac{1}{2t_k} \|x - x^k\|^2 \right\}.$$

Hence

$$P_C(x^k - t_k \nabla f(x^k)) = \arg \min_{x \in C} \left\{ \langle \nabla f(x^k), x \rangle + \frac{1}{2t_k} \|x - x^k\|^2 \right\},$$

□

As a consequence, minimizing the linearized objective with quadratic regularization over a convex set C is equivalent to projecting the gradient step onto C in the Euclidean norm:

$$x^{k+1} = \Pi_C \left(x^k - t_k \nabla f(x^k) \right).$$

The approach is motivated by a general principle in smooth convex optimization: rather than minimizing the original function, one minimizes a local upper bound. When ∇f is L -Lipschitz continuous, the *Descent Lemma* guarantees that

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Projected gradient descent follows by minimizing this surrogate, where the quadratic term arises from Euclidean geometry.

In our setting, where optimization is performed over the probability simplex, working under the Euclidean geometry is suboptimal. To address this, we replace the penalty with a divergence function that better reflects the underlying structure.

Mirror descent generalizes *subgradient gradient descent* idea by replacing the norm-based regularizer with a *distance-like* Bregman divergence tailored to the domain.

Definition 3.3.2 (Bregman Distance). *Let $\omega: \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable and μ -strongly convex function with respect to some norm. The **Bregman distance** associated with ω is defined as:*

$$B_\omega(x, y) = \omega(x) - \omega(y) - \langle \nabla \omega(y), x - y \rangle.$$

This divergence is non-negative and vanishes only when $x = y$. Unlike a true distance, it is not necessarily symmetric and does not satisfy the triangle inequality. It becomes particularly useful when the feasible set C has a non-Euclidean geometry, such as the probability simplex.

Replacing the Gradient Descent's quadratic penalty with a Bregman divergence leads to the mirror descent update:

Definition 3.3.3 (Mirror Descent). *Given a differentiable, strongly convex function ω and a convex set C , the **mirror descent** update at step k with subgradient $g_k \in \partial f(x_k)$ and step size $t_k > 0$ is defined as:*

$$x_{k+1} = \arg \min_{x \in C} \left\{ \langle g_k, x \rangle + \frac{1}{t_k} B_\omega(x, x_k) \right\}.$$

As in the Euclidean case, this minimization admits a dual interpretation. The first-order optimality condition reads:

$$0 \in g_k + \frac{1}{t_k} (\nabla \omega(x_{k+1}) - \nabla \omega(x_k)),$$

which rearranges to:

$$\nabla \omega(x_{k+1}) = \nabla \omega(x_k) - t_k g_k.$$

Thus, the update takes the form:

$$x_{k+1} = (\nabla \omega)^{-1} (\nabla \omega(x_k) - t_k g_k).$$

That is, mirror descent performs a gradient step in the dual coordinates $u = \nabla \omega(x)$, and then maps back to the primal space using the inverse mirror map.

When $\omega(x) = \frac{1}{2} \|x\|^2$, the Bregman distance becomes the squared Euclidean distance, and mirror descent reduces to projected (sub)gradient descent as shown above.

This generalization allows optimization in domains where the geometry is better captured by a non-Euclidean divergence, often resulting in more efficient updates or better-behaved trajectories.

3.4. Exponential Gradient Descent

A special case of the mirror descent we will focus on in this thesis is **exponential gradient descent**, which corresponds to mirror descent on the probability simplex with negative entropy regularization. In this setting, the mirror map is

$$\omega(x) = \sum_i x_i \log x_i,$$

and its gradient is $\nabla \omega(x) = (1 + \log x_i)_i$. The update then becomes:

$$x_{k+1,i} = \frac{x_{k,i} \exp(-t_k g_{k,i})}{\sum_{j=1}^n x_{k,j} \exp(-t_k g_{k,j})}.$$

This multiplicative weights update is especially useful when $x_k \in \Delta_n$, the probability simplex, as it automatically preserves non-negativity and normalization.

Algorithm 1 Exponential Gradient Descent [Jia22]

- | | |
|--|--|
| 1: $p^0 := \frac{1}{n} \mathbf{1}$ | ▷ Initialize with uniform distribution over n elements |
| 2: for $t = 0, 1, \dots, T - 1$ do | ▷ Iterate for T steps |
| 3: $g^t := \nabla f(p^t)$ | ▷ Compute the gradient of the objective at current iterate |
| 4: $w_i^{t+1} := p_i^t \exp(-\eta g_i^t)$ | ▷ Apply multiplicative weights update to each coordinate |
| 5: $p_i^{t+1} := w_i^{t+1} / \ w^{t+1}\ _1$ | ▷ Normalize to ensure $p^{t+1} \in \Delta_n$ |
| 6: end for | |
| 7: $\bar{p} := \frac{1}{T} \sum_{t=0}^{T-1} p^t$ | ▷ Return the average iterate as found solution |
-

3.5. Joint Mirror Descent for Spectral Deconvolution

We now return to our main optimization objective:

$$\min_p \text{UOT} \left(\mu, \sum_i p_i \nu_i \right) = \min_p \min_G F(G, p),$$

where μ is the observed mixture spectrum, ν_i are the known reference components, p is the vector of nonnegative mixture proportions (lying on the probability simplex), and G is the unbalanced optimal transport plan. The function $F(G, p)$ comprises both the transport cost and regularization terms as in 3.1:

$$F(G, p) = \min_{G \geq 0} (\langle G, \mathbf{M} \rangle_F + \lambda \cdot \text{div}_{\text{reg}}(G, \mathbf{c}) + \tau_1 \cdot \text{div}_m(G \mathbf{1}, \mu) + \tau_2 \cdot \text{div}_m(G^\top \mathbf{1}, \sum_i p_i \nu_i)) \quad (3.2)$$

Instead of solving this bilevel problem using alternating optimization or differentiating through the transport solver (as in [DNG⁺23]), we propose to optimize both p and G simultaneously via a *joint mirror descent* algorithm.

3.5.1. Mirror Descent on Product of Simplices

Let us consider the function $f(x, y)$, convex jointly in $x \in \Delta^n$ and $y \in \Delta^m$, where

$$\Delta^n := \left\{ x \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = 1 \right\}, \quad \Delta^m := \left\{ y \in \mathbb{R}^m : y_j \geq 0, \sum_{j=1}^m y_j = 1 \right\}.$$

In our setting, x corresponds to the flattened transport plan G and y corresponds to the mixture weights p .

We define the mirror descent step using the Kullback–Leibler divergence as the Bregman distance:

$$D_{\text{KL}}(x, x') = \sum_{i=1}^n x_i \log \left(\frac{x_i}{x'_i} \right), \quad D_{\text{KL}}(y, y') = \sum_{j=1}^m y_j \log \left(\frac{y_j}{y'_j} \right).$$

3.5.2. Optimization Step

Given current iterates $x^{(k)}, y^{(k)}$, we compute the joint mirror descent step as:

$$\begin{aligned} (x^{(k+1)}, y^{(k+1)}) = \arg \min_{\substack{x \in \Delta^n \\ y \in \Delta^m}} & \left\{ \langle \nabla_x f(x^{(k)}, y^{(k)}), x \rangle + \langle \nabla_y f(x^{(k)}, y^{(k)}), y \rangle \right. \\ & \left. + \frac{1}{\eta} [D_{\text{KL}}(x, x^{(k)}) + D_{\text{KL}}(y, y^{(k)})] \right\} \end{aligned}$$

This corresponds to performing mirror descent independently over each variable with respect to its KL geometry.

The minimization above yields closed-form multiplicative updates:

$$\begin{aligned} x_i^{(k+1)} &= \frac{x_i^{(k)} \exp(-\eta [\nabla_x f(x^{(k)}, y^{(k)})]_i)}{\sum_{l=1}^n x_l^{(k)} \exp(-\eta [\nabla_x f(x^{(k)}, y^{(k)})]_l)}, \\ y_j^{(k+1)} &= \frac{y_j^{(k)} \exp(-\eta [\nabla_y f(x^{(k)}, y^{(k)})]_j)}{\sum_{l=1}^m y_l^{(k)} \exp(-\eta [\nabla_y f(x^{(k)}, y^{(k)})]_l)}. \end{aligned}$$

Chapter 4

Implementation

Our implementation is inspired by the Python Optimal Transport library [FCGea21], which uses the L-BFGS-B algorithm to solve the unbalanced optimal transport (UOT) problem. However, this approach does not preserve the constraint $\sum_{i,j} \gamma_{i,j} = 1$, whereas our method, based on Mirror Descent, does. Moreover, their implementation relies solely on a dense matrix representation, which poses a significant performance issue. Our implementation is available at: <https://github.com/Orzan/UOTjointMD>.

4.1. Optimal Transport for NMR Spectra

In the context of NMR mixture analysis, the optimal transport plan $\gamma \in \mathbb{R}_+^{n \times n}$ represents how signal mass is shifted from a synthetic (estimated) spectrum to the observed one. Each entry γ_{ij} specifies the amount of intensity transported from position y_j to x_i , where both spectra are discretized over a shared chemical shift grid of size n .

The cost matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ encodes pairwise transport costs between grid points, which we define as $M_{ij} = |x_i - y_j|$. Alternative metrics, such as squared ℓ_2 distance, which emphasize more localized transport, may also be worth exploring.

The main computational bottleneck arises from the fact that both $\mathbf{M}$ and γ scale quadratically with n . For high-resolution spectra (e.g., $n \approx 10^4$), working with full dense matrices becomes impractical.

4.2. Sparse representation

Assuming that the two spectra being compared are similar, the transport should occur only over a short distance. This means that only entries close to the diagonal of $\mathbf{M}$ are meaningful. This motivates the restriction of attention to a fixed bandwidth C , beyond which transport is considered negligible, thus producing a theoretical multidagonal cost matrix M as

$$M_{i,j} = \begin{cases} |x_i - y_j| & \text{if } |i - j| < C, \\ \infty & \text{otherwise.} \end{cases}$$

where x_i and y_j are the chemical shift values of each spectrum respectively, and C is the restriction distance. In practice, we represent this structure as a flat vector $[D_i]$, where each D_i corresponds to one of the $2C - 1$ diagonals of $\mathbf{M}$, centered around the main diagonal and spanning the meaningful bandwidth.

Accordingly, we constrain γ to have the same multidagonal structure. Let G be equal to sparse γ and $a = \mu, b = \sum_i p_i \nu_i$, then we calculate the following:

$$\text{reg_kl}(G, c) = \lambda \sum_{i,j} G_{i,j} \log \left(\frac{G_{i,j}}{c_{i,j}} + 10^{-16} \right) + \sum_{i,j} (c_{i,j} - G_{i,j})$$

$$\text{grad_reg_kl}(G, c) = \lambda \log \left(\frac{G}{c} + 10^{-16} \right)$$

$$\text{marg_tv}(G, a, b) = \tau_1 \sum_i \left| \sum_j G_{i,j} - a_i \right| + \tau_2 \sum_j \left| \sum_i G_{i,j} - b_j \right|$$

$$\text{grad_marg_tv}(G, a, b) = \tau_1 \cdot \text{sign}(G\mathbf{1} - a) \otimes \mathbf{1}^T + \tau_2 \cdot \mathbf{1} \otimes \text{sign}(G^T\mathbf{1} - b)^T$$

Thus, we obtain the full objective and its gradient from Equation 3.2 as:

$$f(G, p) = \langle G, \mathbf{M} \rangle_F + \text{reg_kl}(G, c) + \text{marg_tv}(G, a, b)$$

$$\nabla_G f(G, p) = \mathbf{M} + \text{grad_reg_kl}(G, c) + \text{grad_marg_tv}(G, a, b)$$

$$\nabla_p f(G, p) = -\tau_2 \sum_i \text{sign}(G^T\mathbf{1} - b) \nu_i$$

These procedures are implemented directly in terms of multidagonal storage, completely avoiding dense matrix operations. It significantly reduces both memory usage and computational cost. This reduces the memory complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(nC)$, where $C \ll n$.

4.3. Exponential Gradient Updates for Sparse OT

We now apply the exponential gradient descent (3.4) method to our unbalanced optimal transport objective and construct the step of the descent as described in 3.5.2. Let $G^{(k)} \in \mathbb{R}_+^{n \times n}$ denote the current transport plan and $p^{(k)} \in \mathbb{R}^m$ the current weights, where m is the number of components. We compute an element-wise multiplicative update:

$$W_{ij}^{(k+1)} = G_{ij}^{(k)} \cdot \exp \left(-\eta \partial_{G,ij} f(G^{(k)}, p^{(k)}) \right),$$

$$G^{(k+1)} = \frac{W^{(k+1)}}{\sum_{i,j} W_{ij}^{(k+1)}}$$

and

$$q_i^{(k+1)} = p_i^{(k)} \cdot \exp \left(-\eta \partial_{p,i} f(G^{(k)}, p^{(k)}) \right),$$

$$p^{(k+1)} = \frac{q^{(k+1)}}{\sum_i q_i^{(k+1)}}.$$

where $\eta > 0$ is the step size.

Before we can formulate the whole algorithm, we introduce a more convenient starting point for mirror descent. We construct an initial sparse, multidagonal transport plan G_0 using a warm start strategy, described with the following properties:

- Mass is first assigned along the main diagonal (i, i) as $\min(a[i], b[i])$.

- Any remaining mass is then distributed greedily to adjacent entries $(i \pm k)$, where the offset k satisfies $|k| \leq C$, with C denoting the prescribed bandwidth. This ensures that the transport plan remains sparse and localized.

Pseudocode for the algorithm becomes as follows:

Algorithm 2 Exponential Gradient Descent for Sparse OT

```

1:  $p = \frac{1}{n} \mathbf{1}$  ▷ Initialize weights uniformly
2:  $a := \mu, b = \sum_{i=1} p_i \nu_i$  ▷ Initialize the mixture and weighted component intensities
3:  $M$  ▷ Initialize the transport matrix
4:  $G^0$  ▷ Initialize warmstart as described above
5:  $\mathbf{c} := \frac{1}{|G^0|} \mathbf{1}$  ▷ Initialize  $\mathbf{c}$  with uniform distribution for regularization
6: for  $t = 0, 1, \dots, T - 1$  do
7:    $dG^t, dp^t := \nabla f_G(G^t, p^t), \nabla f_p(G^t, p^t)$ 
8:    $W_i^{t+1} := G_i^t \exp(-\eta_G \cdot dG_i^t)$ 
9:    $G_i^{t+1} := W_i^{t+1} / \|W^{t+1}\|_1$ 
10:   $q_i^{t+1} := p_i^t \exp(-\eta_p \cdot dp_i^t)$ 
11:   $p_i^{t+1} := q_i^{t+1} / \|q^{t+1}\|_1$ 
12:   $b = \sum_{i=1} p_i^{t+1} \nu_i$ 
13: end for
14:  $f(G^{T-1}, p^{T-1})$  ▷ Return the final distance

```

4.4. Python Code

The implementation below performs joint mirror descent over both the transport plan G and the mixing proportions p . We introduce two improvements: a decay factor $\gamma \in (0, 1]$ that progressively reduces the learning rate, and a patience mechanism that terminates optimization when no improvement is observed for a fixed number of iterations. These additions help stabilize convergence and improve time efficiency.

```

1  def joint_md(
2      self,
3      eta_G,
4      eta_p,
5      max_iter,
6      tol=1e-6,
7      gamma=1.0,
8      patience=50
9  ):
10     """
11     Parameters:
12         eta_G (float): Learning rate for transport plan updates
13         eta_p (float): Learning rate for mixing proportion updates
14         max_iter (int): Maximum number of iterations
15         tol (float): Convergence tolerance for relative change in objective
16         gamma (float): Learning rate decay factor (applied each iteration)
17         patience (int): Iterations with no improvement before stopping
18
19     Returns:
20         tuple: (G, p) where G is the optimal transport plan as dia_matrix and
21                p is the optimal mixing proportions as np.array
22     """
23     n = len(s_list) # Number of component spectra

```

```

24 p = np.ones(n) / n # Initialize mixture proportions uniformly
25
26 # Flatten initial sparse G_0
27 G_flat = flatten_multidiagonal(self.G0_sparse.data, self.offsets)
28
29 prev_val = None # Previous objective value
30 stalled_iterations = 0 # Counter for early stopping
31
32 for i in range(max_iter):
33     # ----- Mirror update for transport plan G -----
34     val, grad = self.func_sparse(G_flat) # Compute objective and gradient
35     G_flat *= np.exp(-eta_G * grad) # Apply multiplicative update
36     G_flat /= G_flat.sum() # Normalize
37
38     # ----- Mirror update for proportions p -----
39     grad_p = self._compute_p_gradient(G_flat, nus) # Compute Gradient
40     p *= np.exp(-eta_p * grad_p) # Apply multiplicative update
41     p /= p.sum() # Project back to simplex
42
43     # ----- Update target distribution -----
44     self.b = sum(nu_i * p_i for nu_i, p_i in zip(nus, p))
45
46     # ----- Convergence check based on relative improvement -----
47     if prev_val is not None:
48         rel_change = abs(val - prev_val) / max(abs(prev_val), 1e-10)
49         if rel_change < tol:
50             stalled_iterations += 1
51         else:
52             stalled_iterations = 0
53         if stalled_iterations >= patience:
54             break # Stop if no improvement for 'patience' iterations
55
56     prev_val = val
57
58     # ----- Learning rate decay -----
59     eta_G *= gamma
60     eta_p *= gamma
61
62 # Final reconstruction of transport plan
63 G_data = reconstruct_multidiagonal(G_flat, self.offsets, self.m)
64 return dia_matrix((G_data, self.offsets), shape=(self.m, self.n)), p

```

Listing 4.1: Joint mirror descent implementation with decay and patience

4.5. Finding hyperparameters

There is no straightforward method to determine the values of C , λ , τ_1 , and τ_2 , as these parameters vary between experiments. The bandwidth parameter C can be estimated as a percentage of the spectral dimensionality. The parameter λ should be sufficiently large to smooth the geometry of the problem, yet small enough to avoid distorting the final distance. Through empirical testing, we found that a value of λ around 1.5 yields the best results.

The main challenge lies in selecting appropriate values for τ_1 and τ_2 . For each experiment, we generated a heatmap spanning a range of τ_1 and τ_2 values. For each pair (τ_1, τ_2) , we computed the estimation error using the formula:

$$L_1 = \|w_{\text{estimated}} - w_{\text{true}}\|.$$

Below is one of the heatmaps, showing L1 distance for $w_{\text{true}} = [0.091, 0.909]$:

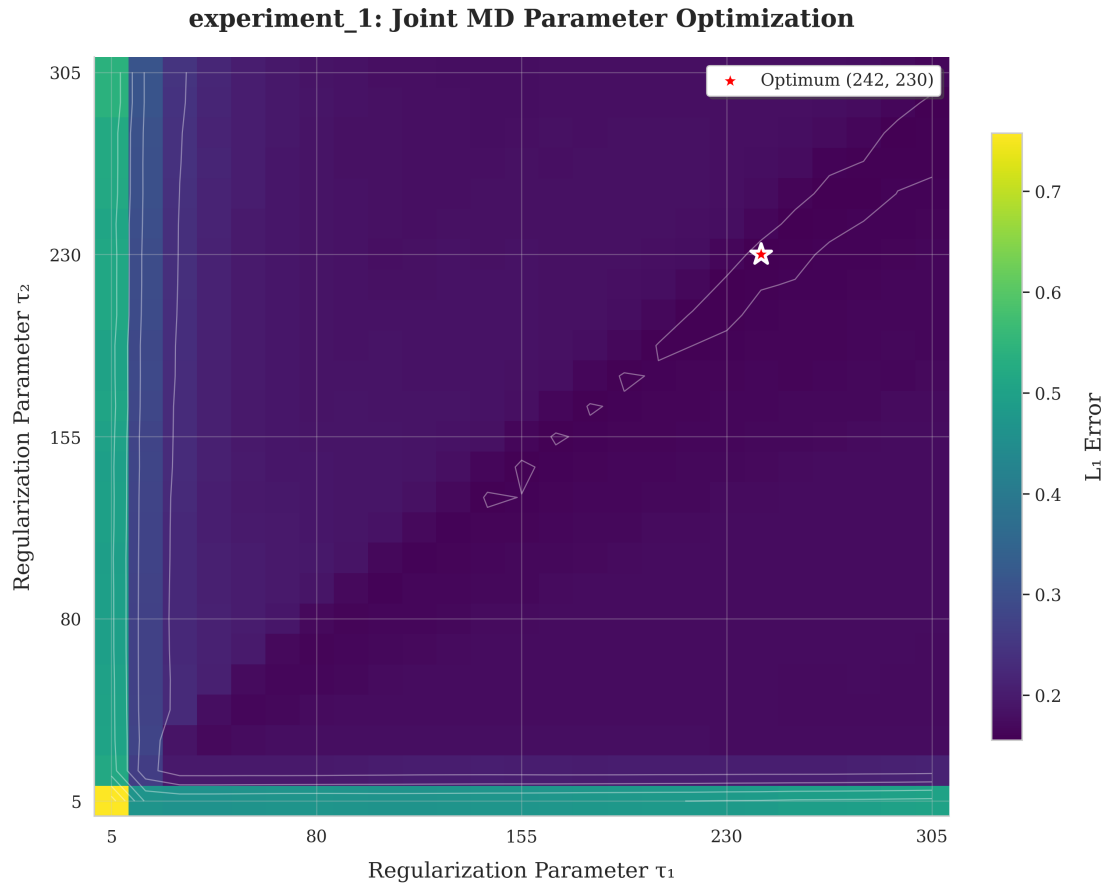


Figure 4.1: Heatmap of L_1 for pairs (τ_1, τ_2)

Chapter 5

Results

5.1. Simple Wasserstein Descent

As a baseline method, which we also used broadly in exploratory research and early experimentation for this thesis, we implemented a simplified approach to spectral deconvolution. This method directly minimizes the Wasserstein distance between the observed mixture and a candidate linear combination of known component spectra. In the one-dimensional setting relevant for NMR spectroscopy, the Wasserstein distance has a convenient closed-form expression:

$$W_p^p(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^p dt,$$

where F_μ^{-1} and F_ν^{-1} denote the quantile functions (inverse cumulative distributions) of the spectra treated as probability measures.

We define the candidate mixture spectrum as

$$\nu_p = \sum_{i=1}^M p_i \nu_i, \quad \text{with } p \in \Delta_M,$$

where Δ_M is the standard probability simplex and ν_i are the reference spectra. The Wasserstein distance between the observed mixture and ν_p is minimized over p using exponential gradient descent:

$$p_i^{(t+1)} \propto p_i^{(t)} \cdot \exp\left(-\eta_t \frac{\partial W_p(\mu, \nu_p)}{\partial p_i}\right), \quad \text{followed by normalization.}$$

This method is implemented in `PyTorch`, allowing efficient differentiation and GPU acceleration. While simple and interpretable, it has important limitations: it lacks regularization mechanisms to handle unmatched mass or noise. As a result, its performance degrades with increasing numbers of components or significant deviations from the convex hull of the reference spectra. Nonetheless, in low-dimensional settings with well-aligned spectra, it provides a fast and competitive baseline that is useful for prototyping and comparison.

5.2. Cross-Experiment Summary

Our proposed Joint Mirror Descent approach outperformed the Simple Wasserstein (Simple WS) baseline in 9 out of 11 benchmark experiments. Table 5.1 and Figure 5.1 summarizes the comparative results.

Table 5.1: Summary of Experimental Results

Exp.	Real proportions	WS proportions	L_{WS}^1	JMD proportions	L_{JMD}^1
1	0.091 0.909	0.079 0.921	0.023	0.095 0.905	0.008
2	0.505 0.495	0.524 0.476	0.038	0.498 0.502	0.013
3	0.726 0.106 0.082 0.086	0.781 0.038 0.150 0.031	0.245	0.725 0.087 0.101 0.086	0.039
4	0.302 0.224 0.125 0.203 0.146	0.233 0.151 0.140 0.151 0.324	0.387	0.209 0.201 0.134 0.211 0.244	0.231
5	0.302 0.224 0.125 0.203 0.146	0.229 0.151 0.131 0.158 0.332	0.384	0.285 0.161 0.017 0.233 0.305	0.378
6	0.386 0.614	0.442 0.558	0.111	0.344 0.656	0.086
7	0.840 0.160	0.900 0.100	0.119	0.846 0.154	0.012
8	0.370 0.630	0.337 0.663	0.067	0.516 0.484	0.292
9	0.302 0.224 0.125 0.203 0.146	0.232 0.154 0.142 0.149 0.324	0.389	0.220 0.194 0.186 0.193 0.207	0.244
10	0.254 0.495 0.251	0.320 0.334 0.346	0.322	0.238 0.494 0.267	0.032
11	0.485 0.243 0.272	0.340 0.325 0.334	0.291	0.517 0.253 0.230	0.084

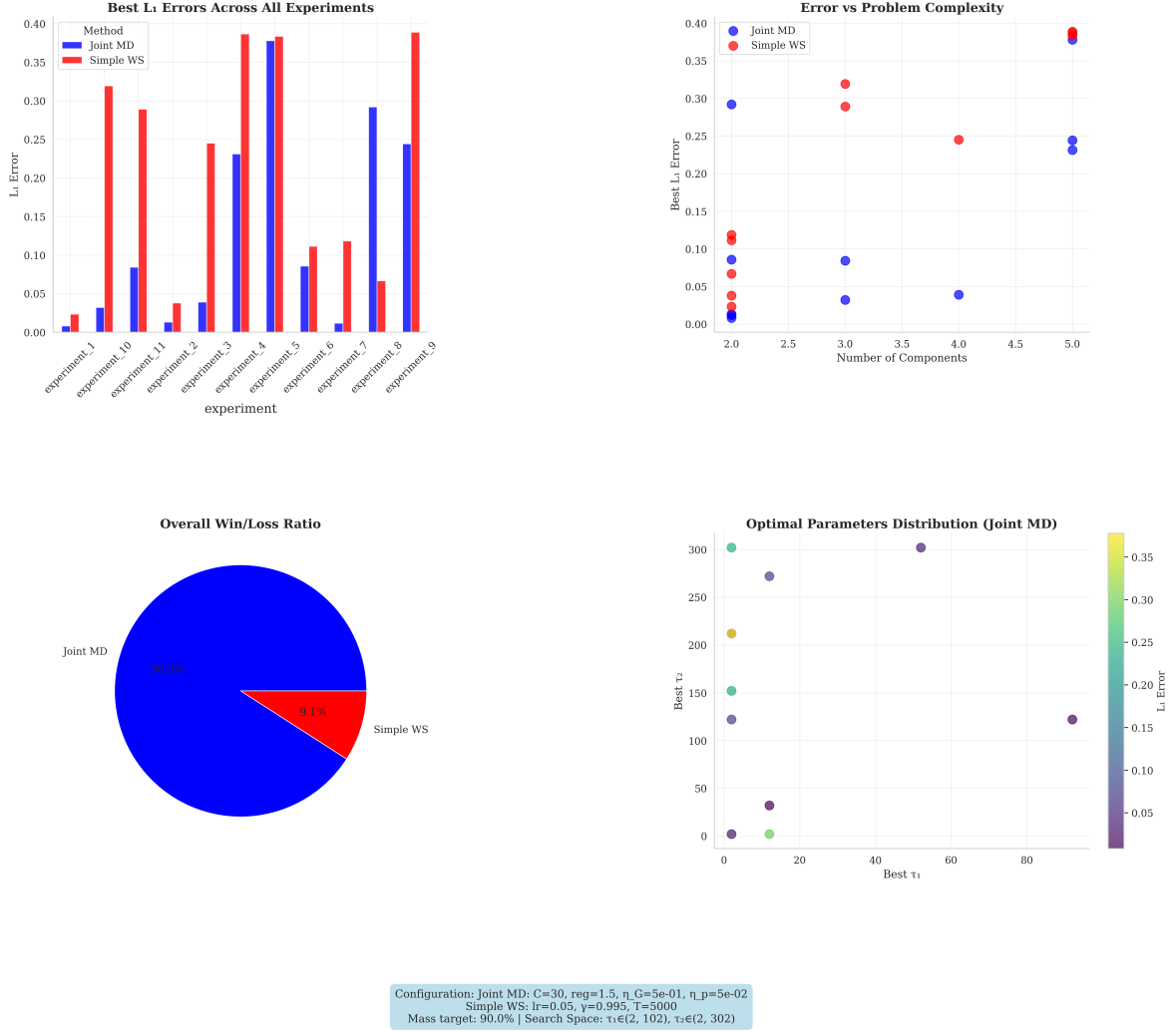


Figure 5.1: Aggregate results across 11 experiments comparing Joint MD and Simple WS. Joint MD wins are marked in blue, Simple WS in red.

For each experiment, we selected the best-performing parameters (τ_1, τ_2) through grid search. The performance advantage of Joint MD grows with spectral complexity (i.e., number of components), suggesting its superior ability to model challenging mixtures. However, we found no single (τ_1, τ_2) configuration that works uniformly well across all experiments, indicating the importance of per-task tuning.

5.3. Parameter Sensitivity and Optimization Landscape

To further understand performance trends, we visualized the objective surface and sensitivity of error with respect to both regularization parameters.

experiment_1: Parameter Optimization Surface

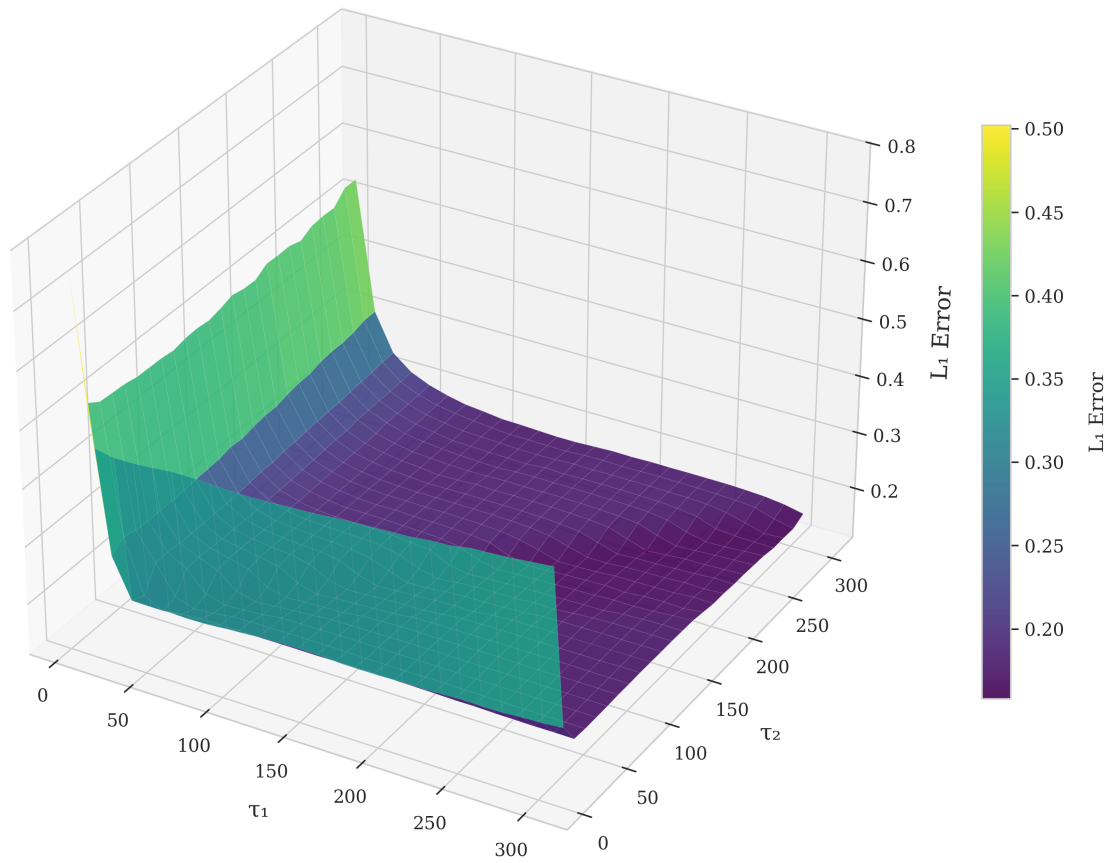


Figure 5.2: 3D error surface for (τ_1, τ_2)

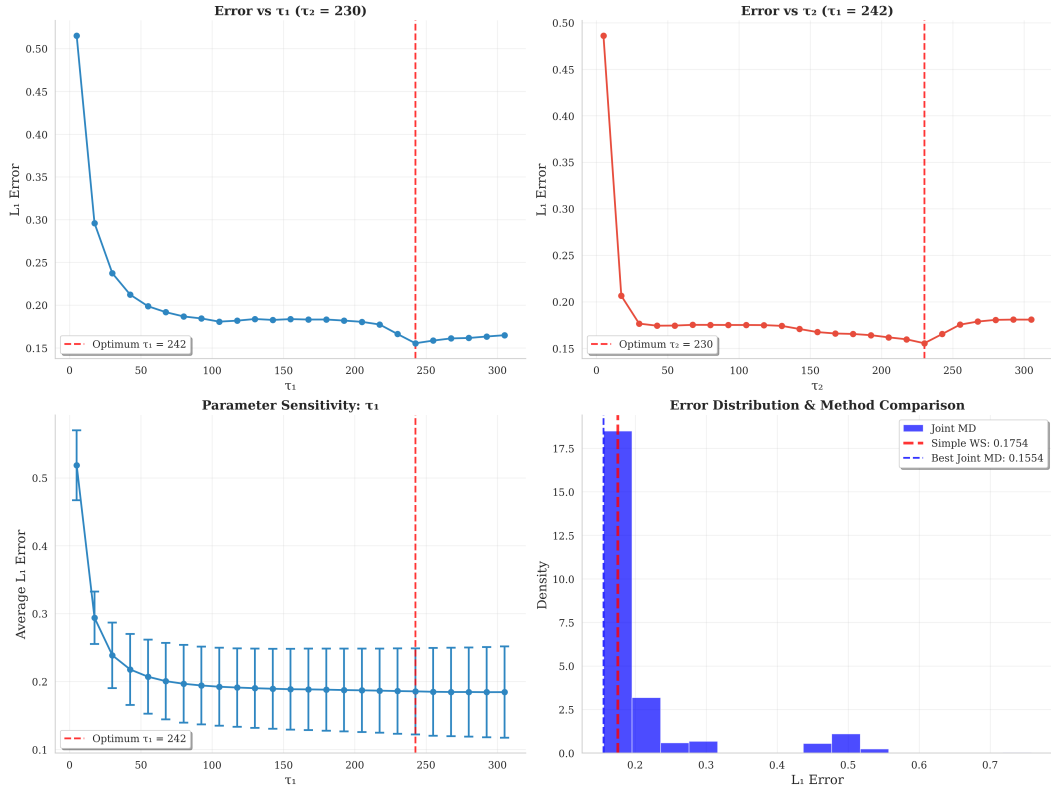


Figure 5.3: Error trends across τ_2 for selected τ_1

These plots reveal a consistent pattern: low regularization ($\tau_1 \approx 0$) yields poor reconstructions, while moderate values offer substantial gains. However, excessive regularization beyond a certain point no longer improves results. The dependence on τ_2 is smoother, with a gentle valley near the optimal range.

5.4. Experiment-Specific Evaluations

We now examine two representative experiments in greater detail to illustrate the relative performance and robustness of our method.

5.4.1. Experiment with Two Components (experiment_1)

As a relatively simple test scenario, **experiment_1** allows for a baseline comparison of methods with limited complexity.

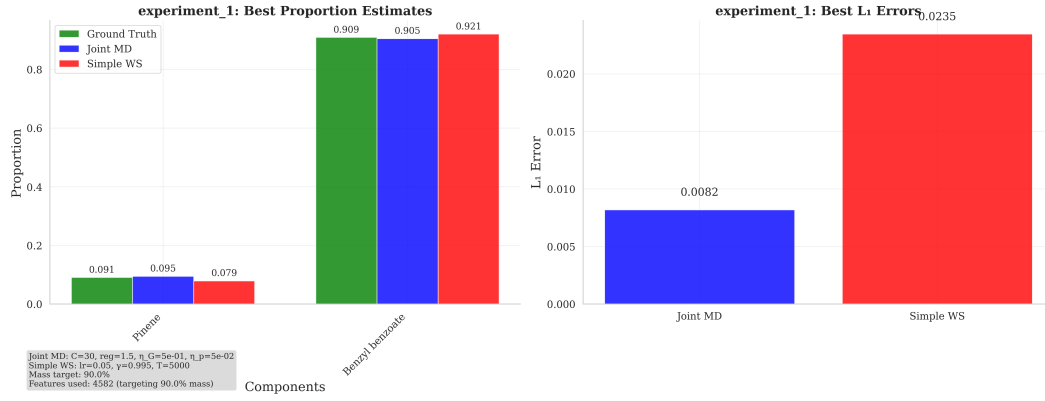


Figure 5.4: Joint MD estimation comparison vs. Simple WS for `experiment_1`.

In this simpler case, Simple WS still achieves competitive performance. Nonetheless, Joint MD provides an improvement in accuracy, highlighting the advantage of our regularized optimization approach even in lower-complexity scenarios.

5.4.2. Complex Experiment with Four Components

In contrast, for the more challenging scenario involving four varied component spectra, our Joint MD method demonstrates significant superiority over the Simple WS baseline.

Figures below illustrate the advantage of joint mirror descent. The optimization surface is sharply defined, indicating a clear region of optimality where the method achieves much better accuracy.

experiment_3: Parameter Optimization Surface

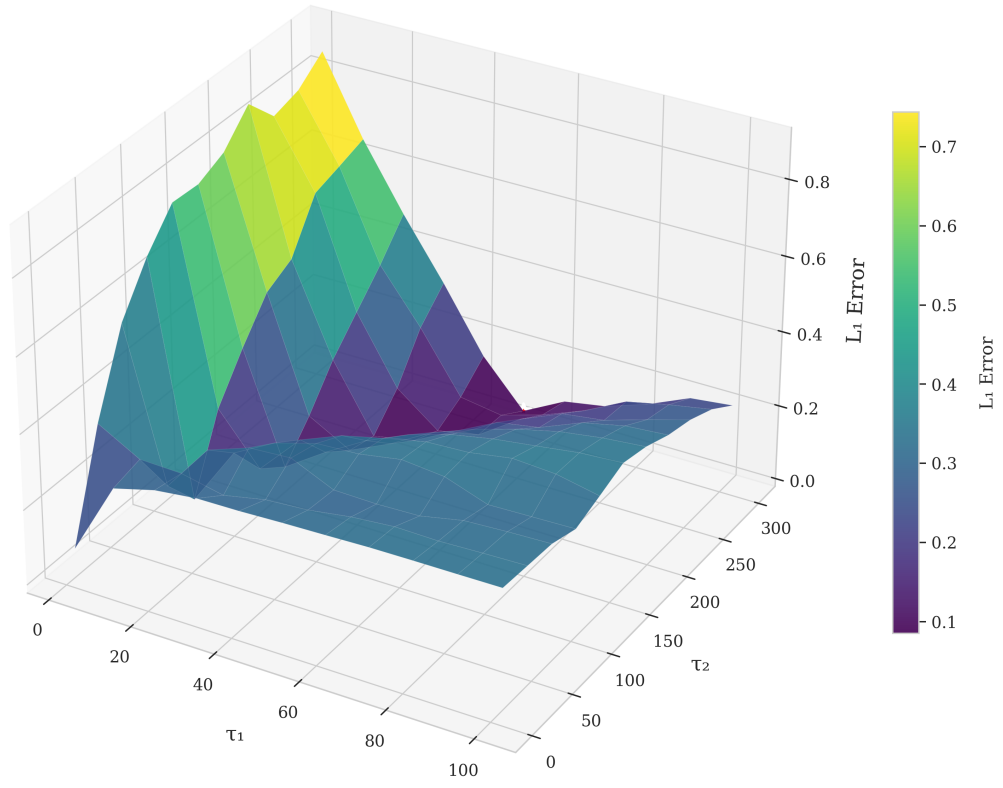


Figure 5.5: Error surface (τ_1, τ_2) for **experiment_3**

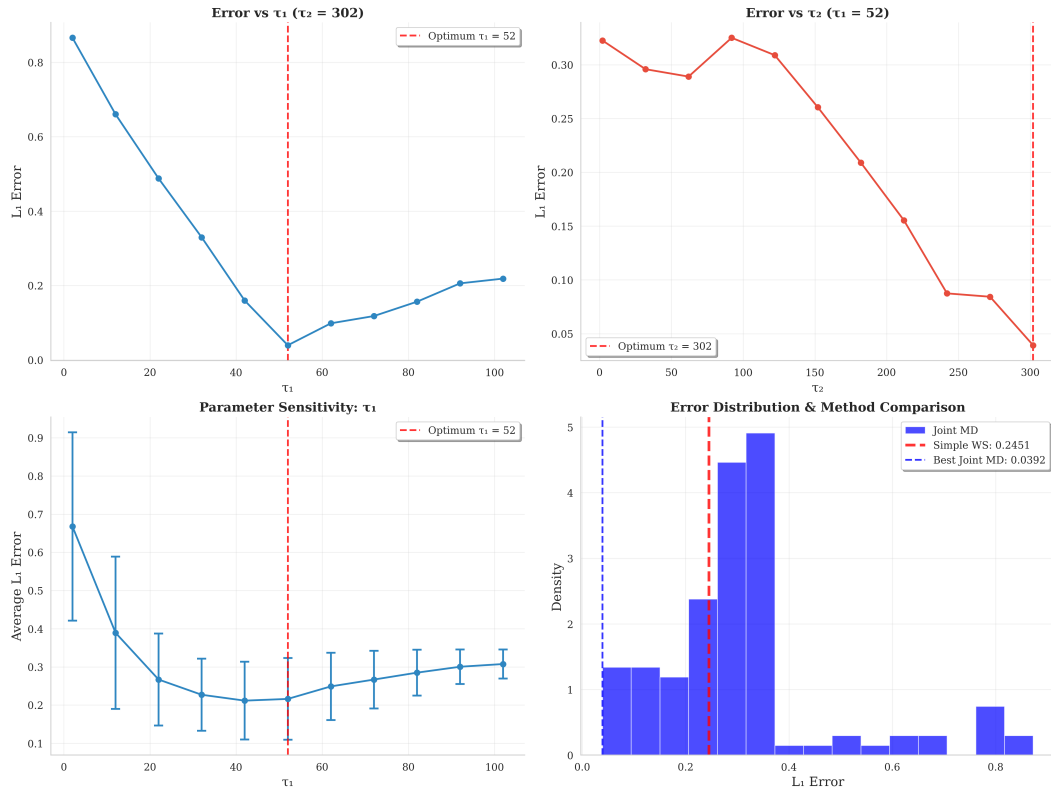


Figure 5.6: Parameter trends for experiment_3

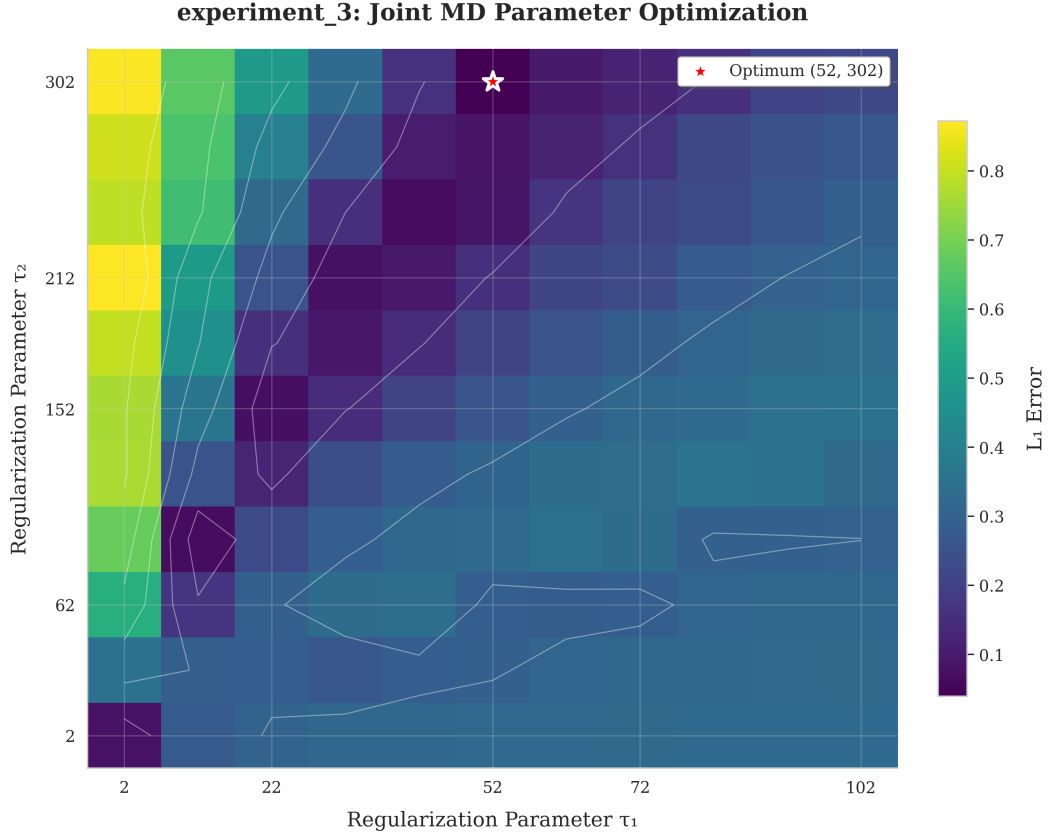


Figure 5.7: Heatmap of L1 error over (τ_1, τ_2) for **experiment_3**

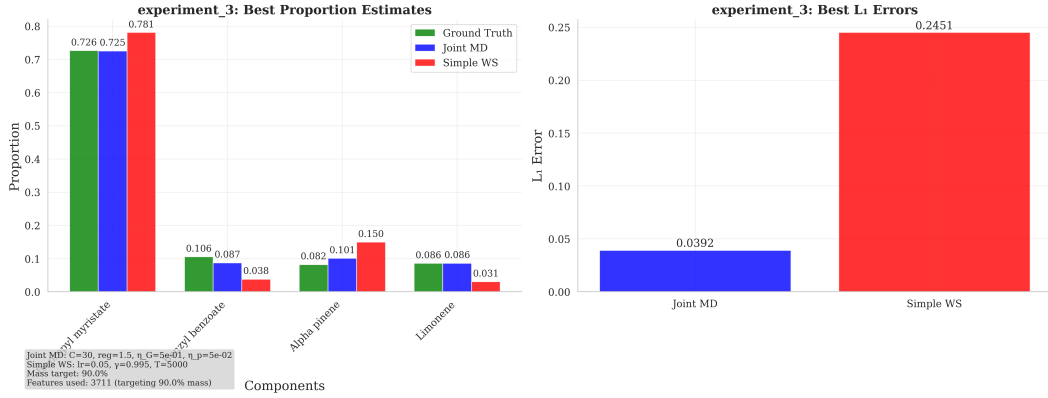


Figure 5.8: Estimations Joint MD vs. Simple WS on **experiment_3**

Overall, these results strongly support the use of the proposed joint mirror descent method as an effective solution for challenging spectral deconvolution tasks, particularly when handling a higher number of complex spectral components.

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